Socioeconomic Effects of AI

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Abstract

This essay delves into the effect of artificial intelligence to our job market. It argues beyond the commonly spread idea of "AI is taking all the jobs" and draws on analysis from the Congressional Budget Office's report on AI's true effect on labor market, feseral budgent and overall economy. This essay studies Philip Moreira Tomei and Bouke Klein Teeselink's Reinforcement Learning Feasibility Index and analyzes MIT economist Daron Acemoglu's projections of artificial intelligence. The essay stresses the importance of how companies handle the implementation of AI. The outcome of AI influence is dependent less on raw AI capability and more so whether companies support their partners in this transition by investing in training rather than layoffs, aiding people adapt to changes rather than just eliminating positions entirely. AI is not destroying or creating jobs, it is reshaping them and the method to our management of this transition is where our economy is to be determined.

Keywords: AI and employment, labor market, economic impact, reinforcement learning feasibility.

“AI is taking our jobs.” We’ve all heard some version of this statement whether it is “AI is taking over”, “AI is taking our jobs” or something else along those lines. While we surf, swim or nearly drown in this wave of a recession it is the only thing we hear when we come up to air. What if I told you this is simply not true.

According to Dan Katz, First Deputy Managing Director of the IMF, in Artificial Intelligence and the Economics of Adjustment, AI is unlikely to permanently reduce employment. He predicts it will instead shift the task composition, skill requirements and organizational structure of workspaces entirely. While AI may change many jobs it will absolutely increase productivity, which can cause economic growth, wages and living standards. Similar to the Congressional Budget Office’s framing in Artificial Intelligence and Its Potential Effects on the Economy and the Federal Budget, I believe artificial intelligence is another technological revolution which will cause massive evolution in terms of society, technology and economy altogether. My own view aligns with the IMF’s: we’re currently in the middle of an adjustment period as society gets introduced to AI.

Katz makes an argument which deeply resonated with me about how whether it is the steam engine, electrification, computing and more which may have displaced jobs they also stimulate a new demand; the Jevons paradox. Katz understands the fear of AI and refers to it as the modern version of “lump-of-labor-fallacy”; mistaken belief that there’s a fixed amount of work to go around, so that whatever AI does, humans do less of. However, history disagrees and we’ve seen new technologies tends to create jobs just in sectors which didn’t exist yet.

A key part of this transition must involve the aid of companies. Rather than laying off large parts of their workforce, I believe companies should hold training sessions, courses, and information sessions around AI usage. We need to build broader knowledge around this technology to make the transition smoother. While AI may eliminate certain tasks, the goal should not be protecting outdated positions, but helping employees and businesses adapt to new opportunities. As the IMF states, 'The goal of policy should not be to protect specific jobs, companies, or industries. It should be to incentivize employees and businesses to adapt.' Through analyzing various articles and incorporating perspectives from both employers and employees, I have identified three key steps to guide a successful AI transition. The first step is building cross-functional teams to explore where AI can help. The second is focusing on building trust with employees. The third is creating dedicated AI-literacy training to improve productivity and workplace relationships.

MIT economist and 2024 Nobel laureate Daron Acemoglu takes a more cautious view. He argues that while AI has real long-term potential, its near-term economic benefits are likely to be more gradual and modest than most forecasts suggest, unless AI gets meaningful better at supporting complex, real-world work. Technology should be redirected toward providing accurate reliable, task specific information which helps solve current world affairs. Organizations from IMF to Goldman Sachs and McKinsey project massive GDP and productivity gains. However, according to Acemoglu's estimations artificial intelligence will only increase the GDP of the United States by approximately 1.1% over the next ten years and productivity by 0.7%. These statistics massively surprised me, and while I understand not all tasks can be handed to AI, I suspected it would at least affect them to a greater extent. Acemoglu argues that although AI may influence nearly 20% of tasks in the U.S. labor market, only approximately 5% are likely to be profitably automated soon. The CBO's own report echoes same note of caution about how little we actually know. CBO cites survey data showing only approximately 5% of U.S. businesses currently rely on AI in goods production and services, and that while 80% of the U.S. workforce could have at least one-tenth of their tasks affected by AI, only 19% would see at least half their tasks affected. Thus while an overall exposure to AI is common a deep disruption to a single job is much more rare.

What Jobs Can AI Learn? Measuring Exposure by Reinforcement Learning takes this a step further, introducing a Reinforcement Learning Feasibility Index designed to estimate which occupations are most likely to be automated as AI's capabilities improve rather than estimating how exposed occupations are to today's large language models alone. Authors Phillip Moreira Tomei and Bouke Klein Teeselink argue that current measures are limited because they focus solely on present-day capabilities and don't account for future improvements. Their framework scored 17,951 individual O*NET tasks across 894 occupations, first classifying whether each required physical or digital work, then scoring the digital tasks and aggregating the results to the occupation level. Prior to scoring, each task was required to pass a physical feasibility gate: tasks requiring real physical embodiment failed automatically receiving zeroes for their scores (this occurred within 41% of all tasks). Tasks which were passed were scored across 8 sectors revolving how verifiable the outcome is, how the environment can be stimulated, and how much the task resembles a clean, gradable decision problem, then aggregated to the occupation level. The results diverge quite sharply from other AI exposure measures. Jobs from musicians, CEOs and microbiologists seem highly exposed to AI overall but score low on RL feasibility since their work verification is quite difficult. Jobs such as gas plant operators to railroad conductors show the opposite: less exposed by typical measures. However, highly RL-feasible since their work involves clear and checkable decisions.

A point I hadn’t fully considered before is how much the cost of implementing AI may outweigh its benefits. Acemoglu also notes that today’s AI performs best on simple, predictable tasks, it is all pattern recognition with comprehensible success metrics. However, many complex jobs such as teachers, electricians etc. vary from situation to situation and require context, judgement and problem-solving skills which current AI lacks. I’d always assumed AI would eventually ‘takeover’ in some far off, futuristic sense, just in the transition period now. However, I did not give the cost of this transition enough thought. Due to artificial intelligence, current novelty costs may be extremely high, however I believe as it becomes more common on the development as well as deployment side the costs will be lower. This cost analysis shows up in PwC’s Responsible AI: Now that AI is part of your business, what does it mean for audits? Even when a task is technically automatable, PwC notes that adopting AI brings real overhead costs that companies must absorb prior to seeing any return. This involves documentation, monitoring and internal review for every new use case.

One question this leaves open is even if AI does not cause mass unemployment, does it still erode something more subtle? How is it going to affect the way corporations work in terms of seniority, wages etc. What will guarantee a job's survival now? That's a much harder question to measure data to answer and perhaps a more important one.

Artificial intelligence isn’t destroying jobs or creating jobs. It is essentially shifting the size and shape of current employment which comes with AI’s economic impacts. It depends less on technology’s raw capability and rather these transition costs are managed adequately, companies invest in trainings over layoffs and whether polices focus on aiding people adjust to this transition. If done right this new level of technology may be difficult in the short run but quite positive in the long run.

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